Two-Sector DMP Model with Continuous Quality Sorting

Based on Mortensen-Pissarides (1994) and Lagos (2006)

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1 Environment

- **Time:** Continuous.
- **Sectors:** $j \in \{H, L\}$ denoting High-tech and Low-tech (non-high-tech). Sector-specific productivity factors are p_H and p_L.
- **Workers:** Unit mass of risk-neutral workers. Workers are heterogeneous in ability a, which is distributed continuously according to a CDF $F(a)$ with density $f(a)$ on support $[a_{\min}, a_{\max}]$.
- **Labor Force:** The total mass of workers with ability a is $f(a)$. Workers discount the future at rate r. Unemployed workers with ability a receive a flow benefit proportional to their ability, $b \cdot a$.
- **Firms:** Large pool of risk-neutral potential firms. A firm can open a vacancy in sector j at a flow cost c_j.

2 Search and Matching

Workers direct their search to a specific sector based on expected utility, but match randomly with firms within that sector.

- **Sector Choice:** Let $\phi_j(a) \in [0, 1]$ be the probability/fraction of unemployed workers of ability a who choose to search in sector $j \in \{H, L\}$.
- **Sector Pools:** The mass of unemployed workers searching in sector j is $u_j = \int \phi_j(a)u(a)da$, where $u(a)$ is the density of unemployed workers of ability a.
- **Market Tightness:** Let v_j be the number of vacancies in sector j. Market tightness is $\theta_j = v_j/u_j$.
- **Contact Rates:** The matching function is $M(u_j, v_j)$. The worker job-finding rate is $\lambda(\theta_j) = M(u_j, v_j)/u_j$. The firm vacancy-filling rate is $q(\theta_j) = M(u_j, v_j)/v_j$.
- **Pool Composition:** Conditional on a firm in sector j meeting a worker, the density of the worker's ability is $\pi_j(a) = (\phi_j(a)u(a))/u_j$.

3 Production and Endogenous Separation

- **Match Quality:** Upon meeting, the firm and worker draw a match-specific idiosyncratic quality shock $x \sim U[0,1]$. The uniform distribution has mean 0.5, density $g(x) = 1$, and CDF $G(x) = x$.
- **Output:** A producing match between a sector j firm and a worker of ability a with shock x yields flow output $y_j(a, x) = p_j a x$.
- **Shocks:** New idiosyncratic shocks x' arrive at Poisson rate δ, drawn from the same uniform distribution $U[0,1]$.
- **Exogenous Separation:** Matches are destroyed for exogenous reasons at rate s.
- **Endogenous Separation:** A match is continued if and only if the joint surplus is non-negative, defining a reservation match quality $R_j(a)$. If $x < R_j(a)$, the match separates.

4 Value Functions

4.1 Workers

The value of unemployment for a worker of ability a searching in sector j is $U_j(a)$:

$$rU_j(a) = ba + \lambda(\theta_j) \int_{R_j(a)}^{1} [W_j(a,x) - U_j(a)]dx \tag{1}$$

The value of employment $W_j(a, x)$ satisfies:

$$rW_j(a,x) = w_j(a,x) + \delta \int_{R_j(a)}^{1} [W_j(a,x') - W_j(a,x)]dx' - (\delta R_j(a) + s)[W_j(a,x) - U_j(a)] \tag{2}$$

Optimal Sorting: Workers choose the sector that maximizes their value:

$$U(a) = \max\{U_H(a), U_L(a)\} \tag{3}$$

4.2 Firms

The value of a vacancy in sector j is V_j:

$$rV_j = -c_j + q(\theta_j) \int \pi_j(a) \int_{R_j(a)}^{1} [J_j(a,x) - V_j]dxda \tag{4}$$

Free entry implies $V_j = 0$ for all active sectors.

The value of a filled job $J_j(a, x)$ satisfies:

$$rJ_j(a,x) = p_j ax - w_j(a,x) + \delta \int_{R_j(a)}^{1} [J_j(a,x') - J_j(a,x)]dx' - (\delta R_j(a) + s)J_j(a,x) \tag{5}$$

5 Surplus and Equilibrium Conditions

Wages are determined by continuous Nash Bargaining with worker bargaining power β:

$$W_j(a,x) - U_j(a) = \beta S_j(a,x) \quad \text{and} \quad J_j(a,x) - V_j = (1-\beta)S_j(a,x) \tag{6}$$

The total match surplus $S_j(a,x) = W_j(a,x) - U_j(a) + J_j(a,x) - V_j$ satisfies:

$$(r+s+\delta)S_j(a,x) = p_j ax - rU_j(a) + \delta \int_{R_j(a)}^{1} S_j(a,x')dx' \tag{7}$$

5.1 Job Destruction (Reservation Quality)

The reservation quality $R_j(a)$ is defined by $S_j(a, R_j(a)) = 0$:

$$p_j a R_j(a) - rU_j(a) + \delta \int_{R_j(a)}^{1} S_j(a,x')dx' = 0 \tag{8}$$

Subtracting (8) from (7), we can express the surplus linearly in x:

$$S_j(a,x) = \frac{p_j a(x - R_j(a))}{r+s+\delta} \tag{9}$$

Substituting (9) into (8) and evaluating the integral over the uniform distribution $x \sim U[0,1]$ yields:

$$p_j a R_j(a) - rU_j(a) + \frac{\delta p_j a}{r+s+\delta}\frac{(1-R_j(a))^2}{2} = 0 \tag{10}$$

5.2 Derivatives of the Surplus Function

Using (9), we can derive the partial derivatives of the surplus $S_j(a,x)$ with respect to match quality x and ability a.

Differentiating with respect to x:

$$\frac{\partial S_j(a,x)}{\partial x} = \frac{p_j a}{r+s+\delta} \tag{11}$$

Differentiating with respect to a:

$$\frac{\partial S_j(a,x)}{\partial a} = \frac{p_j(x - R_j(a)) - p_j a R_j'(a)}{r+s+\delta} \tag{12}$$

Note on proportionality: Using the Nash bargaining solution, the value of unemployment can be written as:

$$rU_j(a) = ba + \lambda(\theta_j)\beta \int_{R_j(a)}^{1} \frac{p_j a(x - R_j(a))}{r+s+\delta}dx = ba + \lambda(\theta_j)\beta \frac{p_j a(1-R_j(a))^2}{2(r+s+\delta)} \tag{13}$$

Equating (13) with the expression for $rU_j(a)$ from (10) shows that a cancels out entirely from the equation determining $R_j(a)$. Therefore, $R_j(a) = R_j$ is independent of ability a, which implies $R_j'(a) = 0$.

Consequently, the partial derivative with respect to a from (12) simplifies to:

$$\frac{\partial S_j(a,x)}{\partial a} = \frac{p_j(x - R_j)}{r+s+\delta} \tag{14}$$

5.3 Job Creation (Free Entry)

Imposing $V_j = 0$ yields the job creation condition for sector j:

$$\frac{c_j}{q(\theta_j)} = (1-\beta)\int \pi_j(a)\int_{R_j(a)}^{1} S_j(a,x)dxda = (1-\beta)\int \pi_j(a)\frac{p_j a(1-R_j(a))^2}{2(r+s+\delta)}da \tag{15}$$

5.4 Steady-State Flow Equations

Let $e_j(a)$ be the density of employed workers of ability a in sector j.

$$u(a) + e_H(a) + e_L(a) = f(a) \tag{16}$$

The flow into employment in sector j must equal the flow out:

$$\lambda(\theta_j)\phi_j(a)u(a)(1-R_j(a)) = e_j(a)(\delta R_j(a) + s) \tag{17}$$